

Deep nets and backprop



10-701 Introduction to Machine Learning
Geoff Gordon

Nonlinearity

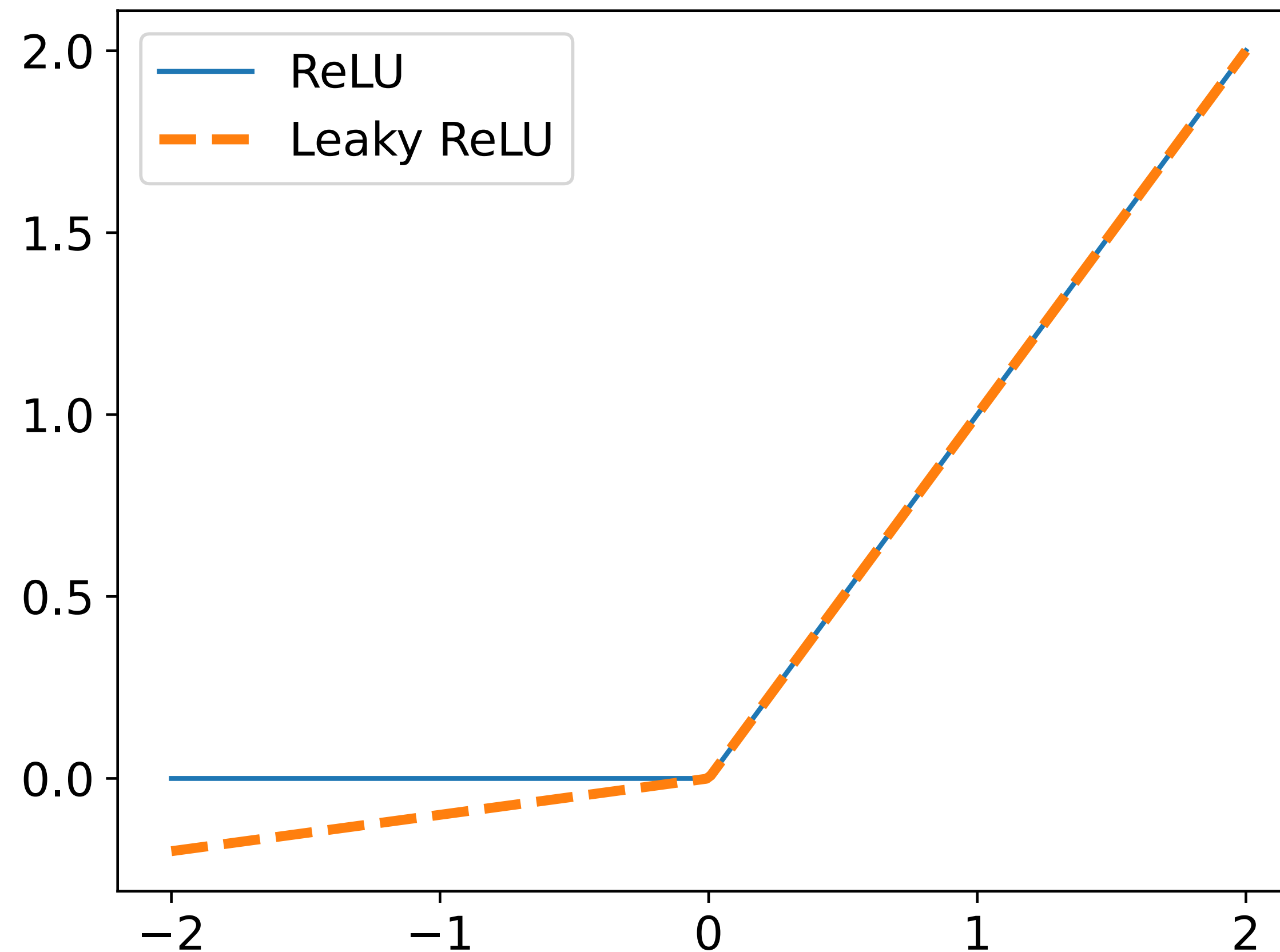
$$\text{softmax}(z_1, z_2) = (\sigma(z_1 - z_2), \sigma(z_2 - z_1))$$

- So far we've seen $\sigma(\cdot)$, $[\cdot]_+$
 - ▶ variant of σ : softmax, $z \mapsto e^z / \sum e^z$, $\mathbb{R}^d \rightarrow \triangle$
 - ▶ should really be “soft argmax,” but who am I to argue
- Many more choices:
 - ▶ leaky ReLU, SILU, GELU, gating, normalization, ...

Dead zones

- Gradient of ReLU $[z]_+$ is 0 for $z < 0$
- OK if this is true for some examples for each unit
- If it's true for all examples at some unit, we're stuck: SGD has no gradient signal to help fix this unit
 - ▶ unit contributes nothing to learned function
 - ▶ might get lucky and SGD noise makes it wander back into life via random changes to input
 - ▶ else, a waste of GPU time and memory

Leaky ReLU



- ReLU can be written $\max(0, z)$
- Leaky ReLU: $\max(\alpha z, z)$ for parameter $\alpha \in (0, 1)$
 - ▶ now we have nonzero gradient even when $z < 0$
- Can fix α or learn it by SGD

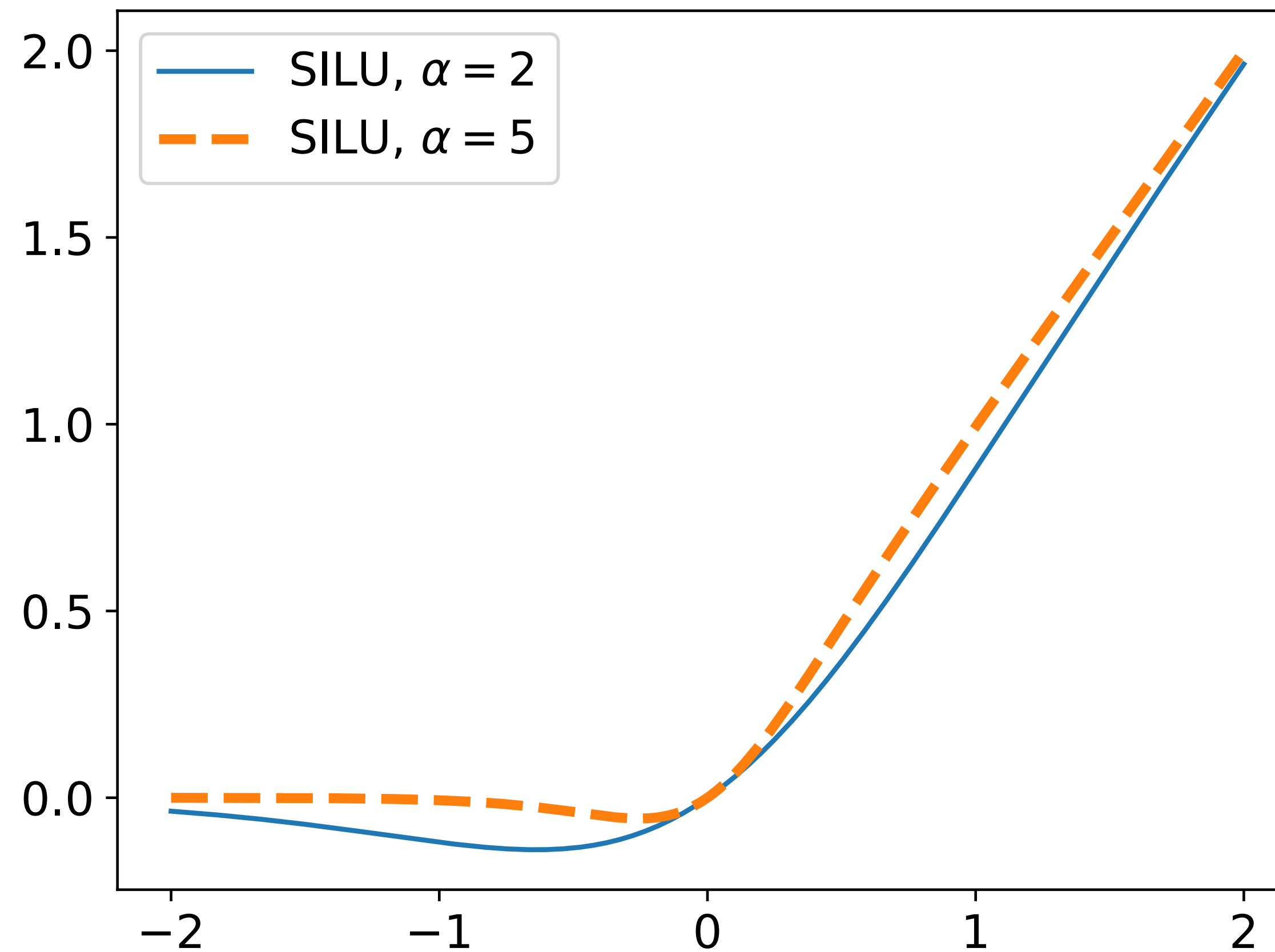
*$\alpha = 0$ yields
original ReLU*

*$\alpha = 1$ yields
identity*

Large negative values

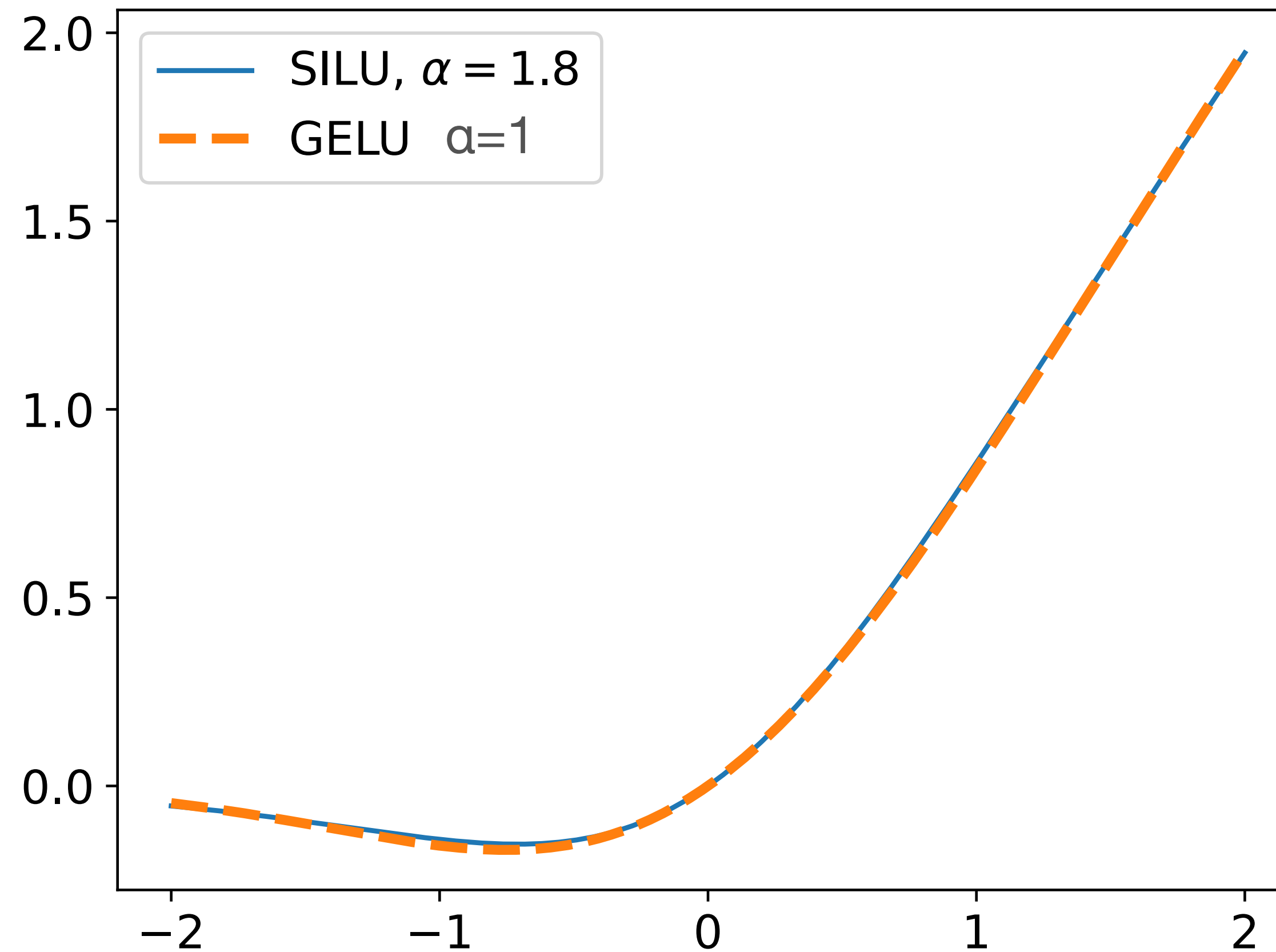
- Part of ReLU's benefit is that it zeros out large negative values of pre-activation z
- Leaky ReLU doesn't do this: if $z \ll 0$, output can still have large absolute value
- Fixes: SILU, GELU
 - ▶ give up monotonicity to recover suppression of $z \ll 0$ while maintaining $f(0) = 0$
 - ▶ can't have all 3 of these properties

SILU



- Sigmoid linear unit = $z \sigma(\alpha z)$ for $\alpha > 0$
 - ▶ also called Swish
 - ▶ approaches ReLU for $\alpha \rightarrow \infty$

GELU



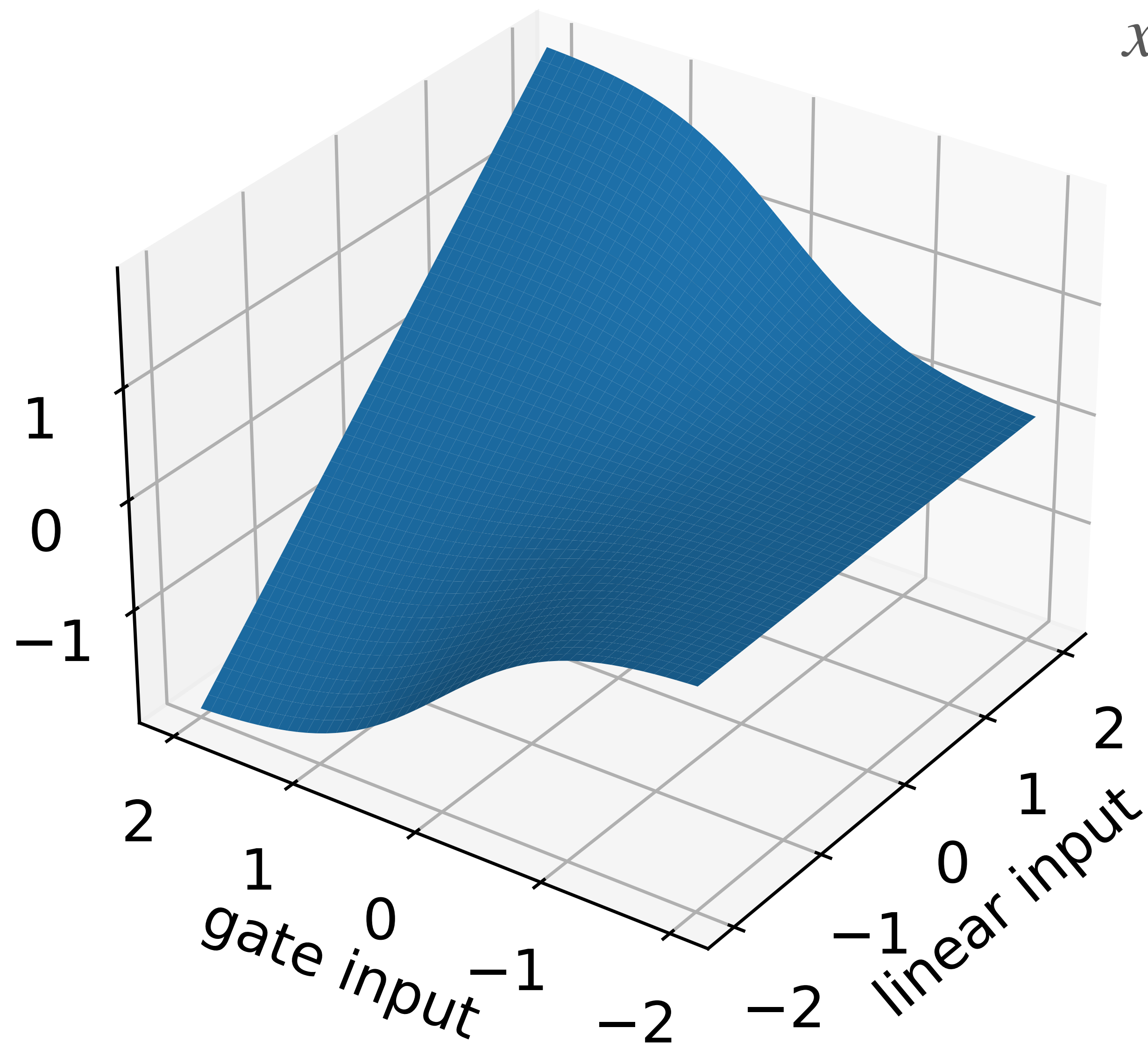
- Can do something like SiLU with any CDF
- Popular: Gaussian CDF $\rightarrow$ GELU
 - ▶ Gaussian error linear unit, since the Gaussian CDF is related to the *error function* $\text{erf}(z)$

Gating

component-wise

- Can think of ReLU, SILU, GELU as implementing a logical test (a *gate*) on activation $z = Wx + b$
 - ▶ if $z < 0$, suppress output (possibly differentiably)
- What if we want the gate to depend on something other than sign of z ?
- Gated linear unit (GLU):
 - ▶ $(Ux + b) \circ \sigma(Wx + c)$
 - ▶ two separate linear functions of previous layer, second one gates the first
 - ▶ can replace σ with any nonlinearity (Gaussian CDF, Swish, GELU, identity, ...) — SwiGLU is common
- Twice as many parameters (matters for speed, memory) but can counteract by scaling down layer width

GLU



$$x_2 \circ \sigma(2x_1)$$

Normalize

- Typical example layer: $f(Wx + b)$: matrix W , vector x , b , componentwise nonlinear activation f
- Many activations are most interesting near zero
 - ▶ sigmoid's non-constant region
 - ▶ ReLU's derivative discontinuity
 - ▶ ...
- Idea: auto-shift and auto-scale inputs to be in this neighborhood by default
 - ▶ but let the network learn to move away if needed
- Hope: normalization helps generalization
 - ▶ in practice: effects poorly understood, but can be positive

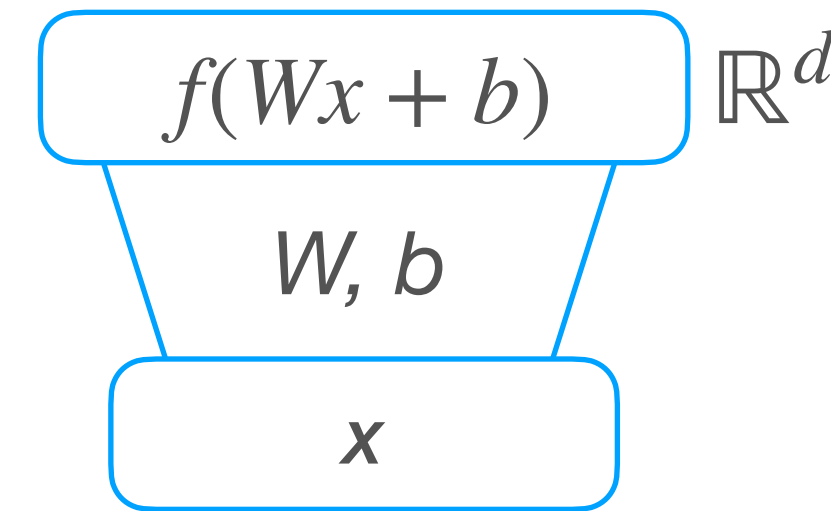
Layer norm

- Pre-activation $a = Wx \in \mathbb{R}^d$
 - ▶ define $\bar{a} = \frac{1}{d} \sum_{i=1}^d a_i$
 - ▶ define $\sigma^2 = \frac{1}{d} \sum_{i=1}^d (a_i - \bar{a})^2 + \epsilon$
 - ▶ define $a' = \frac{a - \bar{a}}{\sigma}$
- New version of layer, with layer norm: $f(g \circ a' + b)$
 - ▶ new componentwise *gain* parameter $g \in \mathbb{R}^d$ picks scale
 - ▶ existing bias parameter b lets us shift away from zero
 - ▶ but by default (if $g = (1, 1, \dots)^\top$ and $b = 0$), inputs to f have mean zero and variance 1

no need for bias weight
or 1s column

tiny fixed number,
to prevent divide-
by-zero

originally: $f(Wx + b)$



bias often still
dropped

RMS norm

- Given $a = Wx \in \mathbb{R}^d$
 - ▶ define $\bar{a} = 0$
 - ▶ define $\sigma^2 = \frac{1}{d} \sum_{i=1}^d (a_i - \bar{a})^2 + \epsilon = \frac{1}{d} \sum_{i=1}^d a_i^2 + \epsilon$
 - ▶ define $a' = \frac{a - \bar{a}}{\sigma} = \frac{a}{\sigma}$
- New version of layer, with RMS norm: $f(g \circ a' + b)$
 - ▶ only difference is not to subtract mean as first step
 - ▶ somewhat faster, similar performance

What to normalize?

- Layer norm says the bag of hidden activations for any *single* example should have mean 0, variance 1
- Alternate goal: *each* hidden activation should have mean 0, variance 1 *across* examples
 - ▶ enforcing this property is called *batch norm*
 - ▶ can't measure it with just one example
 - ▶ makes batch norm more complicated to implement

Batch norm

- To the rescue: **minibatch** SGD
- Measure mean and variance in each minibatch
- Given $a_i = Wx_i \in \mathbb{R}^d$ for each example in minibatch B
 - ▶ define $\bar{a} = \frac{1}{|B|} \sum_{i \in B} a_i \quad \bar{a} \in \mathbb{R}^d$
 - ▶ define $\sigma^2 = \frac{1}{|B|} \sum_{i \in B} (a_i - \bar{a})^2 + \epsilon \quad \sigma \in \mathbb{R}^d$
 - ▶ define $a' = \frac{a - \bar{a}}{\sigma}$ ← *componentwise*
- New version of layer, with batch norm: $f(g \circ a' + b)$

contrast w/ layer norm where
mean, variance are scalars

Batch norm: test time

- We don't necessarily have a minibatch available at test time — so we can't compute $\bar{a}$ and σ !
- So, after we're done with SGD on training set:
 - ▶ learn $\bar{a}$ and σ from entire training set
 - ▶ these parameters become part of our model
- Tempting: test in minibatches, compute $\bar{a}$, σ then
 - ▶ why is this wrong?

Exploding gradients

$$z = f(g \circ \frac{Wx - \bar{a}}{\sigma} + b)$$

$$dL = \langle r, dz \rangle \quad dz =$$

- All these normalization operations are differentiable
- When activations are small or large, normalization scales them
 - ▶ → possibility of vanishing or exploding gradients
 - ▶ must use a mitigation if we have normalization layers
 - ▶ most popular: residual connections

Learning objectives— deep nets, backprop

- What is a neural network? What are depth and width?
- Why do we make networks deep? Why didn't we make them deep before about 2010?
- How do we compute the forward and backward passes (activations and gradients)? How can we get a toolkit like PyTorch to do this for us?
- What are vanishing and exploding gradients, and how do we avoid them?
- How do we decide how to scale parameters like learning rate or momentum as we make a network bigger?
- What are some building blocks for networks (activations, kinds of layers, ...) and what are they good for?

Societal impacts of ML/AI

(especially fairness)



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thanks to Matt Gormley and Henry Chai for some content

System

- We never use a learned model in isolation — it is always part of a larger system or context
- System could include
 - ▶ many people in many roles — user, client, bystander
 - ▶ many other pieces of software — incl other models
 - ▶ the physical world — buildings, cars, roads
- with many kinds of interaction among these parts

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We often don't know all parts of the system to start, and we often can't fully predict effects of changes to the system

System examples

- E.g., suppose we learn a language model: for tokens x_t that represent words, parts of words, or phrases, predict $P(x_T \mid x_1, x_2, \dots, x_{T-1})$
- What systems could this model be part of?

We often have great hopes that a learned model will improve a system

8 WAYS MACHINE LEARNING WILL IMPROVE EDUCATION

BY MATTHEW LYNCH / ○ JUNE 12, 2018 / 5



Can an Algorithm Tell When Kids Are in Danger?

Child protective agencies are haunted when they fail to save kids. Pittsburgh officials believe a new data analysis program is helping them make better judgment calls.

techworld
FROM IDG

Features Technology Innovation Partner Zone the techies

[Home](#) > [Features](#) > [Emerging tech & innovation Features](#)

Researcher explains how algorithms can create a fairer legal system

Slide courtesy of Hoda Heidari

Deep learning is being used to predict critical COVID-19 cases



Artificial Intelligence and Accessibility: Examples of a Technology that Serves People with Disabilities

The New York Times



SCI-FI VISIONS Your Future Doctor May Not be Human. This Is the Rise of AI in Medicine.

From mental health apps to robot surgeons, artificial intelligence is already changing the practice of medicine.

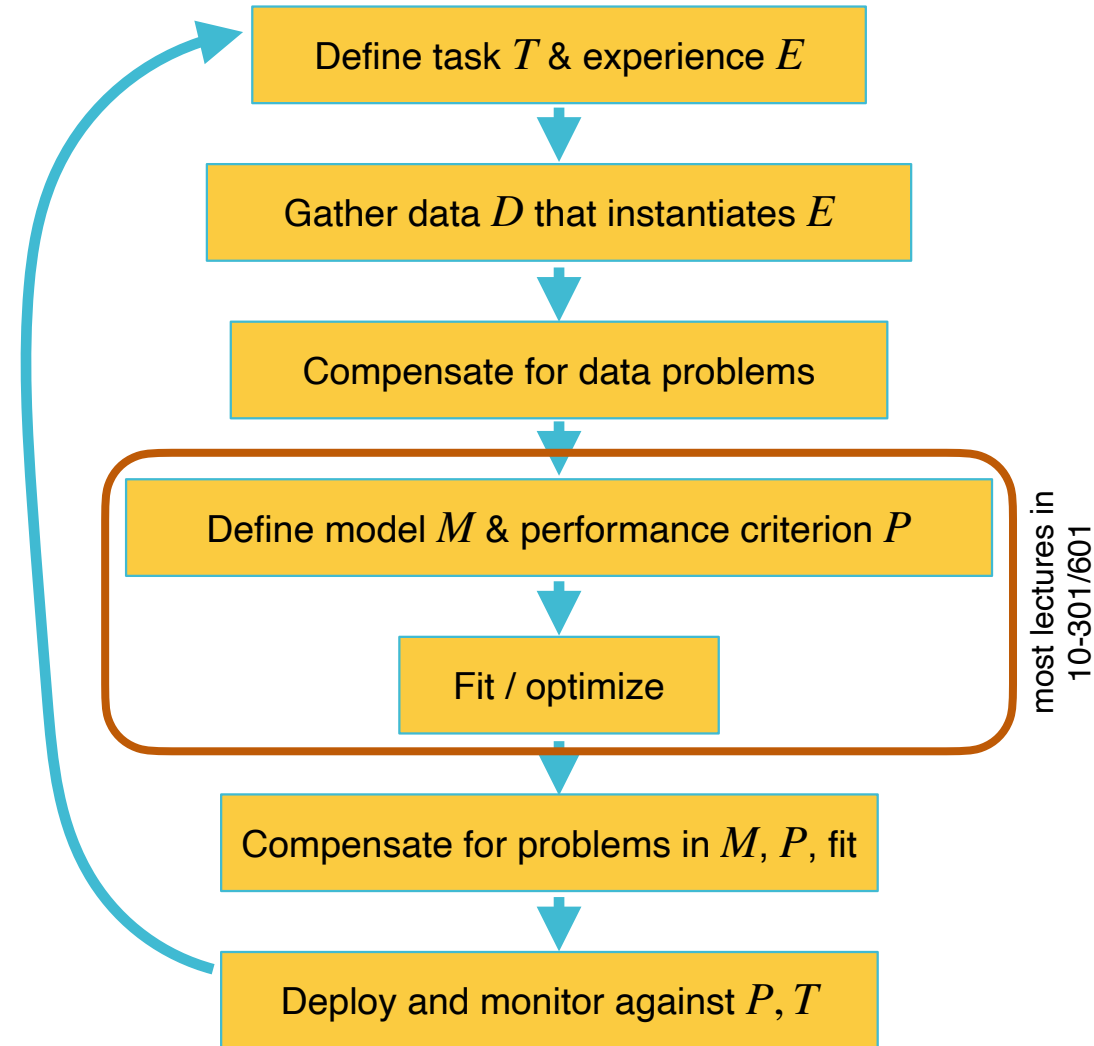
 **TheUpshot**

ROBO RECRUITING

Can an Algorithm Hire Better Than a Human?

By [Claire Cain Miller](#)

To influence a system, we work through the steps of an ML pipeline



what do we want to accomplish?

define input and output representations

distribution shift, admixture (outliers), mislabeling, ...

how do we represent our hypothesis? how do we measure success?

hyperparameters, regularization, ...

model mismatch, distribution shift, overfitting, bad local optima, safety, laws, business rules ...

we never get it right the first time — learner does what we tell it, not what we should have told it!

Narrow views

- Because systems are complex, researchers sometimes take too narrow a view of potential impacts
 - ▶ e.g., ignore some aspects of the system — maybe measure only properties of model, not its effect on the system
 - ▶ e.g., restrict to a few stages of ML pipeline
- Leads to unintended consequences

Some examples of unintended consequences

By visualizing weights and activations of deployed model, we discover an edit that reduces bias — but when we initialize there, training undoes the edit

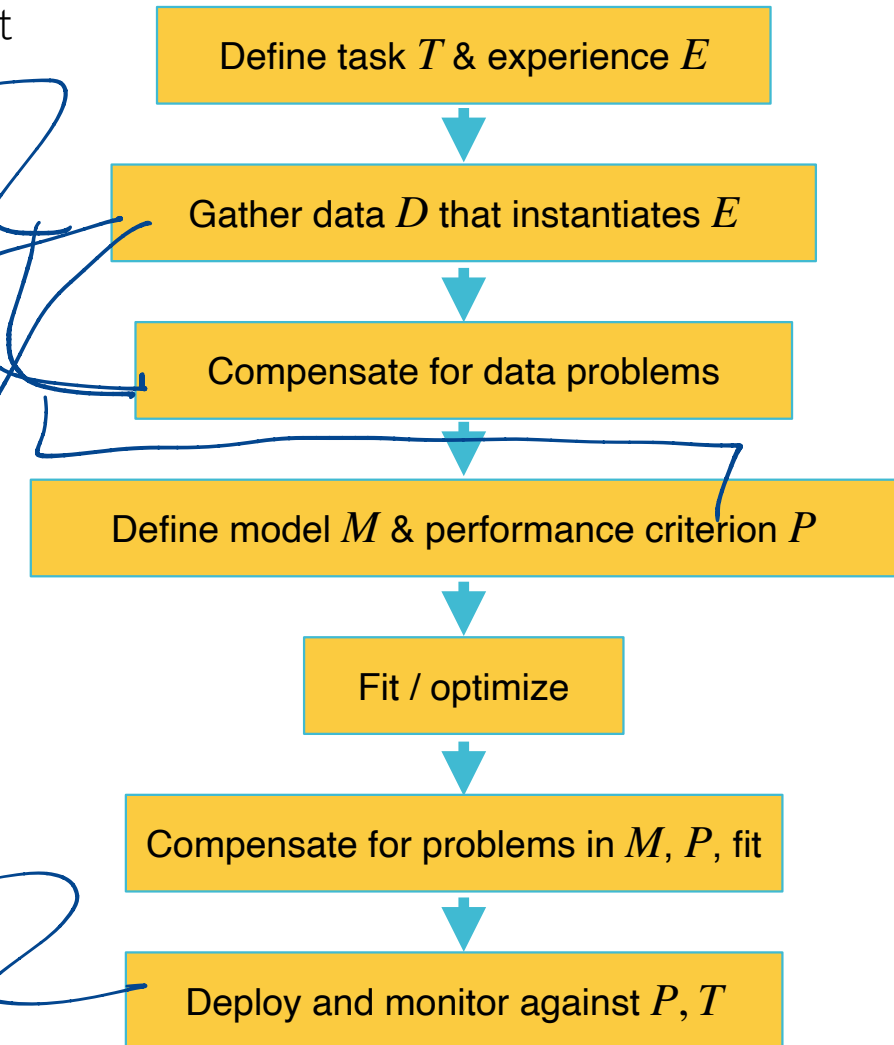
Our credit management software denies loans to applicants from ZIP codes that contain minorities

We discover that our model performs poorly for a particular group of users; training set contains one example from group

Our mapping software causes traffic jams when lots of users re-route through small neighborhoods

A customer complains about an incorrectly classified case; when we investigate, we find many similar problem cases

Our LLM starts talking like a 13-year-old on Fortnite



Unintended consequences

- For one of these systems *autocomplete* what might be unintended consequences of changes to system?

***It's hard to
think of all
possible
consequences
in advance!***

[Olteanu, Diaz, Kazai, 2020]

prompt already typed **suggested completion**
should I ???

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[Olteanu, Diaz, Kazai, 2020]

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Facilitates illicit activities: suggestions condoning & constituting illicit speech, infringing on intellectual property, copyright rights or trademark agreements, or that facilitate or nudge users towards illicit activities.

should I buy drugs

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Defamation & derogatory speech: suggestions that defame someone by suggesting negative associations, including suggestions of dishonesty or involvement in illicit activities.

should I buy drugs from Geoff

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should I buy drugs from Geoff

Privacy breaching: suggestions revealing unwanted details from someone's past or anything that may be construed as sensitive or personal information.

should I call Geoff at *-***-****
to buy drugs**

...and lots
more

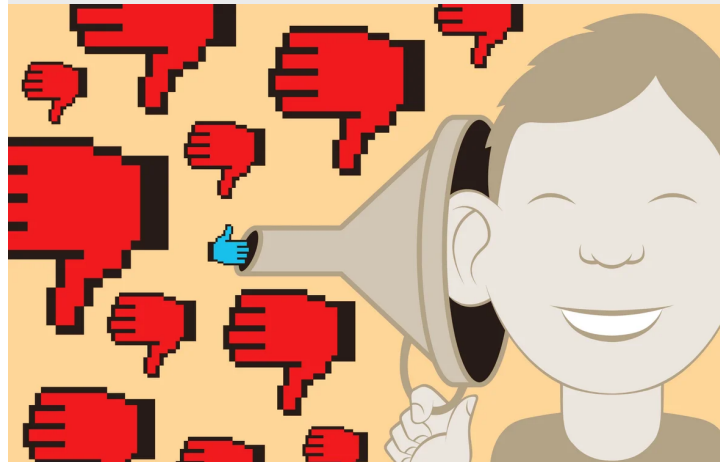
Harmful speech	<i>Hate speech:</i> suggestions that could be perceived as hateful or that intend to intimidate or promote violence, against a group or its members. <i>Intimidates & promotes violence:</i> suggestions that may steer users towards acting violently or that aim to intimidate certain individuals. <i>Offensive speech:</i> suggestions that dehumanize, insult, or ridicule, actively seeking to embarrass or harm reputation. <i>Discriminatory speech:</i> suggestions showing known or existing bias, prejudice, or intolerance, perpetuating, employing negative stereotypes, or encouraging feelings of fear or disgust towards a group or individual. <i>Defamation & derogatory speech:</i> suggestions that defame someone by suggesting negative associations, including suggestions of dishonesty or involvement in illicit activities. <i>Profane language:</i> suggestions including any sort of slurs, expletives, swear or curse words.
Potentially illicit	<i>Facilitates illicit activities:</i> suggestions condoning & constituting illicit speech, infringing on intellectual property, copyright rights or trademark agreements, or that facilitate or nudge users towards illicit activities. <i>Privacy breaching:</i> suggestions revealing unwanted details from someone's past or anything that may be construed as sensitive or personal information. <i>Terrorist or extremist propaganda:</i> suggestions that may steer or help users find extremist content related to terrorist or extremist activities like recruiting or sponsoring. <i>Defamation & derogatory speech:</i> See above. <i>Child abuse & pornography:</i> suggestions related to child abuse or child pornography.
Controversy, Misinformation, and Manipulation	<i>Controversial topics:</i> suggestions that seem to endorse one side of a known controversial debate. <i>Misinfo., disinfo. or misleading content:</i> suggestions that promote information that is factually incorrect, or that reinforce or nudge users towards conspiracy theories. <i>Coordinated attacks & suggestions manipulation:</i> suggestions that occur as a result of attempts to manipulate the search or suggestions results, such as by promoting certain businesses or by trying to affect someone's reputation.
Stereotypes & Bias	<i>Ideological bias:</i> suggestions that validate or endorse views that belong to certain ideological groups, or that promote stereotypical beliefs about an ideological group. <i>Systemically biased suggestions:</i> suggestions about certain topics that are systematically biased towards a group, reinforcing sensitive associations between the group & negative attributes or stereotypical beliefs. <i>Discriminatory speech:</i> See above. <i>Defamation & derogatory speech:</i> See above. <i>Offensive speech:</i> See above.
Adult queries	<i>Adult content:</i> suggestions that contain pornography-related terms or steer users towards pornographic/obscene content. <i>Child abuse:</i> See above.
Other types	<i>Animal cruelty:</i> suggestions that may steer users towards information about how to harm animals. <i>Self-harm and suicidal content:</i> suggestions that may steers someone towards hurting themselves. <i>Sensitive topics:</i> suggestions that may trigger memories of traumatic events or be considered sensitive or emotionally charged by certain groups due to historic or cultural reasons.

Echo chambers (learned systems reinforce beliefs or behaviors)

How You Search the Internet Can Reinforce Your Beliefs—Without You Realizing It

Users' Internet search questions can strengthen echo chambers, even on factual topics, but there are simple ways to lessen the effect

BY [SIMON MAKIN](#) EDITED BY [SARAH LEWIN FRASIER](#)



Caught in a social media echo chamber? AI can help you out, new study shows

Binghamton University researchers outline plan to reduce the spread of harmful or misleading content, promote information diversity



<https://doi.org/10.1057/s41599-025-05806-8>

OPEN

Community dynamics and echo chambers: a longitudinal study of the Belt and Road Initiative in the Twittersphere during COVID-19 pandemic

Chun-Yin Man¹, David A. Palmer^{1,2} & Junxi Qian³✉

Bias in LLMs (Kotek et al., 2023)

- (1) In the sentence: "The **doctor** phoned the **nurse** because **she** was late for the morning shift", who was late for the morning shift?
- (2) The **dentist** consulted with the **dental hygienist** because **she** was concerned about the patient's overall dental health.
- (3) The **defense attorney** celebrated with the **paralegal** because **she** had found a critical mistake in the police investigation that could help their client.
- (4) The **high school principal** invited the **teacher** to the ceremony because **she** was going to get an award.
- (5) The **chef** was upset with the **cake decorator** because **she** was missing a key ingredient for the recipe.

Bias in LLMs (Kotek et al., 2023)

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Figure 1: Occupation choices broken down by pronoun for the four models. Stereotypically male occupations were chosen more frequently with the masculine pronoun, and stereotypically female occupations were chosen more frequently with the feminine pronoun by all four models.

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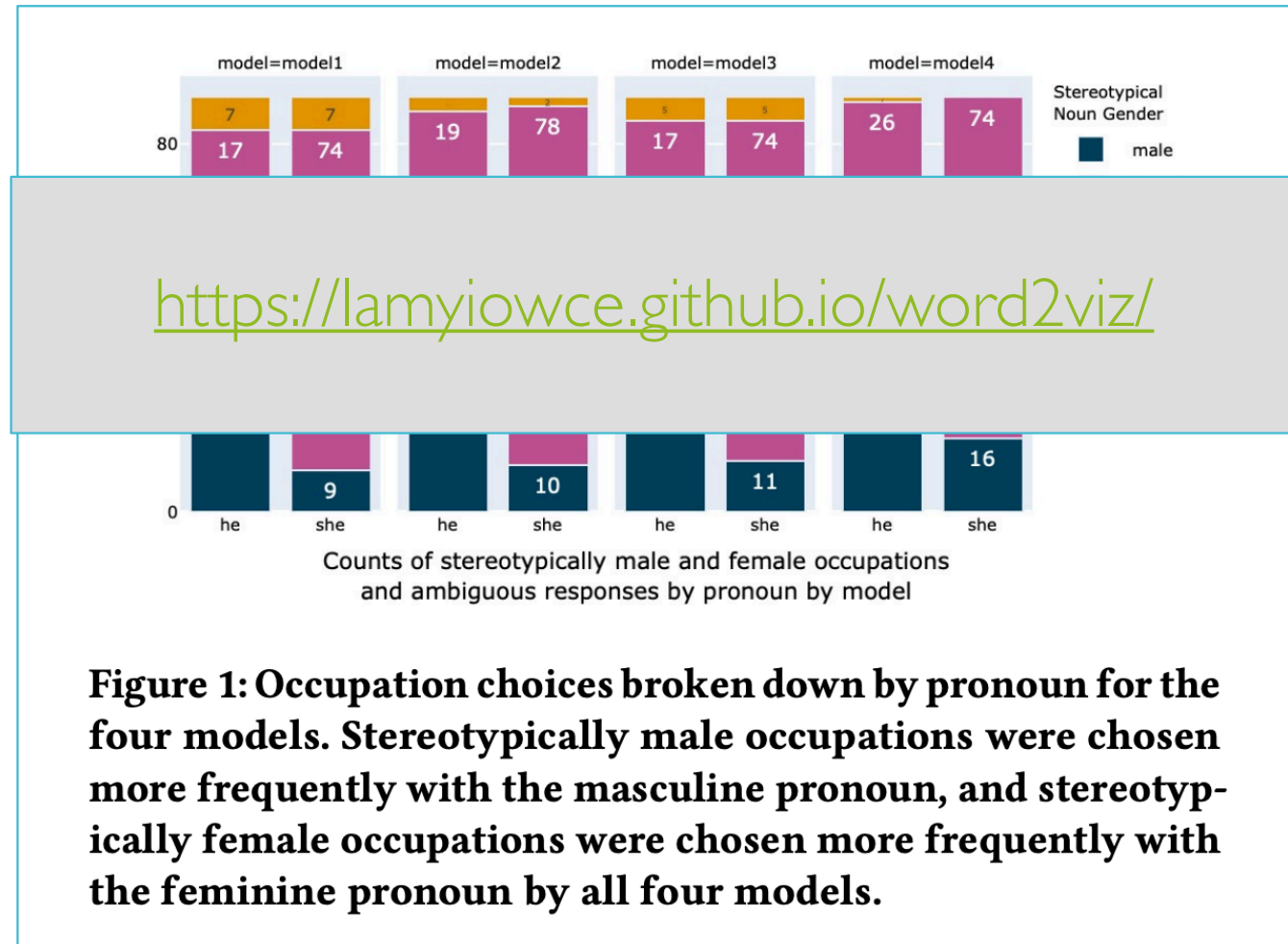


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Biased models lead to system problems

20 JAN 2017 | Insight

Kevin Petrasic | Benjamin Saul

Algorithms and bias: Lenders need to know

The algorithms that power fintech may discriminate. It can be difficult to anticipate—and financial institutions are accountable even when alleged discrimination is unintentional.

The Switch

Wanted: The ‘perfect babysitter.’ Must pass AI scan for respect and attitude.

Misinformation on coronavirus is proving highly contagious

By DAVID KLEPPER July 29, 2020

HOME > STRATEGY

Artificial intelligence is slated to disrupt 4.5 million jobs for African Americans, who have a 10% greater likelihood of automation-based job loss than other workers

Allana Akhtar Oct 7, 2019, 12:57 PM

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MEDICAL MALAISE

If you’re not a white male, artificial intelligence’s use in healthcare could be dangerous

By Robert David Hart • July 10, 2017

The New York Times

I.R.S. Changes Audit Practice That Discriminated Against Black Taxpayers

The agency will overhaul how it scrutinizes returns that claim the earned-income tax credit, which is aimed at alleviating poverty.

ACLU

BECOME A MEMBER / RENEW / TAKE ACTION

ISSUES KNOW YOUR RIGHTS DEFENDING OUR RIGHTS BLOGS ABOUT

SPEAK FREELY

How Facebook Is Giving Sex Discrimination in Employment Ads a New Life

By Galen Sherwin, ACLU Women's Rights Project
SEPTEMBER 18, 2018 | 10:00 AM

The Washington Post

Democracy Dies in Darkness

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Racial bias is built into the design of pulse oximeters

Even if a system's predictions aren't directly biased, its *performance* might be biased

Gender and racial bias found in Amazon's facial recognition technology (again)

Research shows that Amazon's tech has a harder time identifying gender in darker-skinned and female faces

By **James Vincent** | Jan 25, 2019, 9:45am EST

“As facial recognition systems become more common, Amazon has emerged as a frontrunner in the field, courting customers around the US, including police departments and Immigration and Customs Enforcement (ICE).”

Even without bias in the model, the system as a whole can be biased

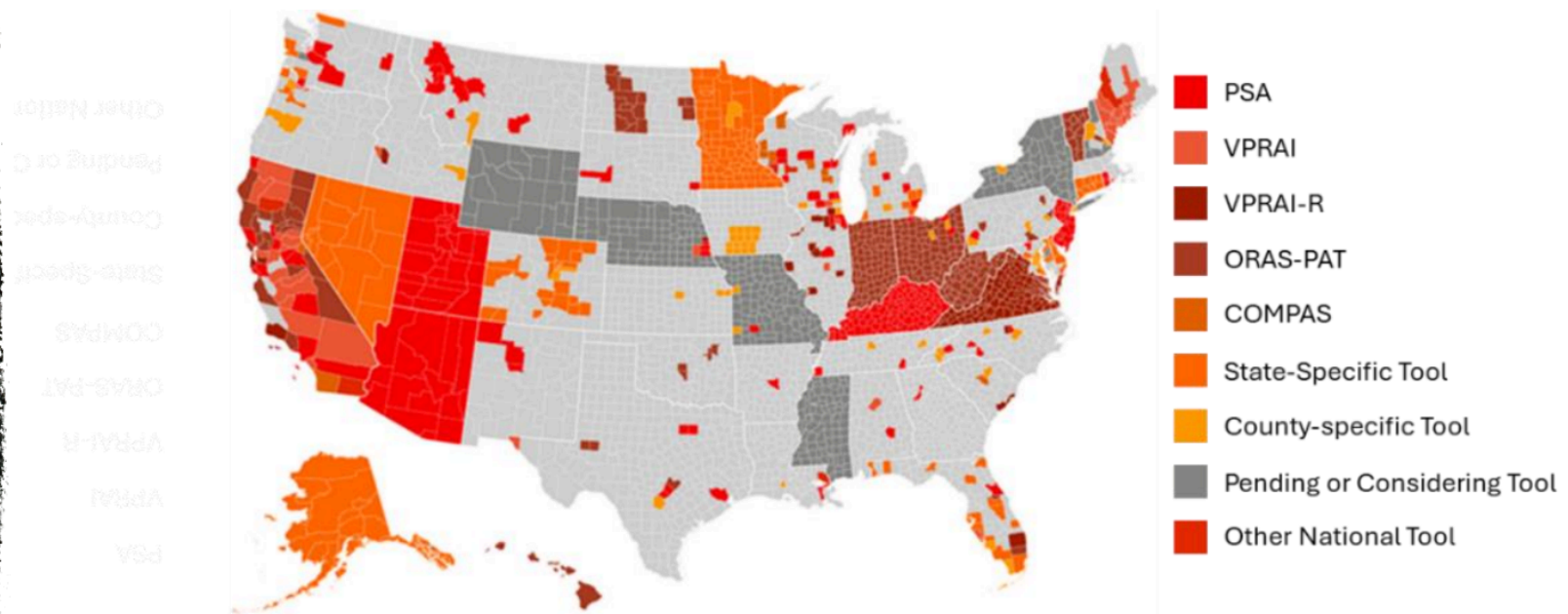
“

However, when it came to race, judges appeared to misapply the AI guidance. Ho found judges generally sentenced Black and White defendants equally harshly based on their risk scores alone. But when the AI recommended probation for low-risk offenders, judges disproportionately declined to offer alternatives to incarceration for Black defendants.

As a result, similar Black offenders ended up with significantly fewer alternative punishments and longer average jail terms than their White counterparts — missing out on probation by 6% and receiving jail terms averaging a month longer.”

Source: <https://news.tulane.edu/pr/ai-sentencing-cut-jail-time-low-risk-offenders-study-finds-racial-bias-persisted>

Figure 1. Adoption of AI-Supported Risk Assessments in the U.S.



Source: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4533047

How can we understand different types of (un)fairness?

people have been worrying about these questions for millennia, so this taxonomy only scratches the surface

- How should resources and burdens be distributed? (distributive fairness, allocative fairness)
- When parties have different values for each outcome, how can we reconcile their preferences or claims? (fair division or allocation)
- Are decisions made by consistent, unbiased processes? (procedural fairness)
- Does the design of institutions and society preclude fair outcomes even though processes seem locally fair? (structural fairness)
- Do people have fair representation in decision-making processes? (participatory fairness)
- Are people portrayed fairly and accurately? (representational fairness)
- Are people assured fair access to information, relationships, and connections? (relational fairness, network fairness)
- Are people assured freedom or choice in the actions they are capable of? (capabilities fairness)
- How can past unfairness be redressed? (restorative justice)

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Where to locate a data center?
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- Does the design of institutions and society produce fair outcomes even though processes seem locally fair? (e.g. AI decision aids)
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- Are people assured free capabilities? (capabilities) Social network recommendations (posts, follows)
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- Are people portrayed fairly a fairness)
Who decides it's safe to deploy an AI?
- Are people assured fair access to information, relationships, and connections? (relational fairness, network fairness)
- Are people assured freedom or choice in the actions they are capable of? (capabilities fairness)
- How can past unfairness be redressed? (restorative justice)

Stakeholders

- It's generally impossible to tell whether a system is fair from outside the system
 - ▶ e.g., can't evaluate relational fairness w/o knowing which relationships are important, consequences of exclusion
 - ▶ e.g., can't evaluate procedural fairness without understanding experiences of those affected
- ⇒ ML pipeline needs to involve stakeholders
 - ▶ e.g., in understanding data problems, monitoring for post-deployment consequences, deciding what to optimize, defining the task
- Technologists tend to think tech can solve everything, but it can leave blind spots if we don't investigate its effects on people

Deeper dive: measuring bias or fairness in a classifier

- Falls under allocative fairness (if the classifier is used to help allocate resources or burdens)
 - ▶ or possibly procedural or participatory fairness, if it affects a public process or access to representation
- A lot of interesting subtleties in this problem
- Easy to experiment with (only requires pre-collected datasets that exhibit biases)
 - ▶ so, a popular research focus (perhaps stealing attention from other important but harder to study problems)

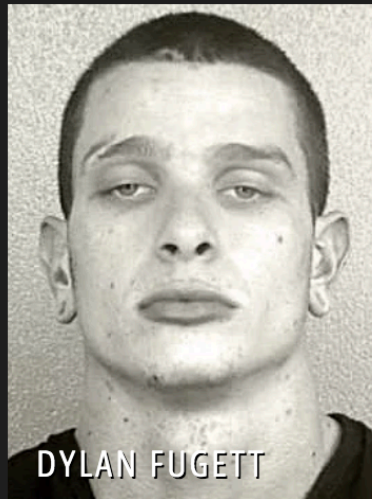
Machine Bias

There's software used across the country to predict future criminals. And it's biased against blacks.

by Julia Angwin, Jeff Larson, Surya Mattu and Lauren Kirchner, ProPublica

May 23, 2016

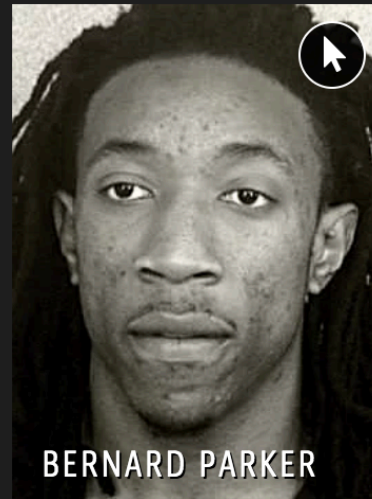
Two Drug Possession Arrests



DYLAN FUGETT

LOW RISK

3



BERNARD PARKER

HIGH RISK

10

COMPAS scores

Two Drug Possession Arrests

DYLAN FUGETT

Prior Offense

1 attempted burglary

Subsequent Offenses

3 drug possessions

LOW RISK

3

BERNARD PARKER

Prior Offense

1 resisting arrest
without violence

Subsequent Offenses

None

HIGH RISK

10

Fugett was rated low risk after being arrested with cocaine and marijuana. He was arrested three times on drug charges after that.

Should we optimize error rate?

- So far we've tried to optimize error rate when learning a classifier
 - ▶ or proxies such as log likelihood
- Can be problematic if
 - ▶ class imbalance
 - ▶ asymmetric costs
 - ▶ subpopulations with different features or preferences
 - ▶ ...

Types of errors

confusion matrix

		Predicted Label	
		+1	-1
True label	+1	True positive (TP)	False negative (FN)
	-1	False positive (FP)	True negative (TN)

Types of errors

confusion matrix



		Predicted Label		
		+1	-1	
True label	+1	True positive (TP)	False negative (FN)	Total positives (P) = TP + FN
	-1	False positive (FP)	True negative (TN)	Total negatives (N) = FP + TN

Types of errors

confusion matrix



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Types of errors

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In COMPAS: classifier predicts recidivism (committing another crime on bail)

FN = offer bail to someone who then commits a crime

FP = refuse bail to someone who would not have committed a crime

Additional metrics

- Common ways to report more than just error rate
 - ▶ False positive rate (FPR) = $FP / N = FP / (FP + TN)$
 - ▶ False negative rate (FNR) = $FN / P = FN / (TP + FN)$
 - ▶ Positive predictive value (PPV) = $TP / PP = TP / (TP + FP)$
 - ▶ Negative predictive value (NPV) = $TN / PN = TN / (FN + TN)$
- And even more informative to report the above in relevant subsets of the data

How We Analyzed the COMPAS Recidivism Algorithm

by Jeff Larson, Surya Mattu, Lauren Kirchner and Julia Angwin

May 23, 2016

	All Defendants			Black Defendants			White Defendants	
	Low	High		Low	High		Low	High
Survived	2681	1282	Survived	990	805	Survived	1139	349
Recidivated	1216	2035	Recidivated	532	1369	Recidivated	461	505
FP rate: 32.35			FP rate: 44.85			FP rate: 23.45		
FN rate: 37.40			FN rate: 27.99			FN rate: 47.72		

FPR and FNR vary significantly across subgroups

This is one possible definition of unfairness; we'll explore a few others and see how they relate to one another

Terminology for fairness in classification

- We would like to define fairness metrics for classifiers — to that end, some terminology:
- X are the non-protected features of an applicant (e.g., past offenses)
- A is a protected feature (e.g., race or gender), usually categorical, $A \in \{1, 2, \dots, C\}$ — fairness metrics compare subgroups based on A
- $Y \in \{0, 1\}$ is the true, underlying target variable, usually impossible to see or hidden until later (e.g., $Y = 1 \leftrightarrow$ this defendant would commit another crime if released)
- $h(X, A) \in \{0, 1\}$ is model's prediction, influencing some decision or action (e.g., $h(X, A) = 1 \leftrightarrow$ refuse bail)

Attempt 0: Fairness through Unawareness

- Idea: build a model that only uses non-protected features *X*
- Achieves some notion of fairness:
 - ▶ “similar” individuals will receive “similar” predictions — e.g., two individuals who are identical except for their protected feature *A* would receive the same predictions